ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

Q1(b) [3 marks]

ADC step = 5/4096 = 1.2207 mV. I divide by 0.1 mV/N, giving ΔF = 12.207 N.

Q1(c) [3 marks]

Using forward Euler, F approximately 1 + AT = 0.8 and G approximately BT = 0.1. Hence x[k+1] approximately 0.8x[k] + 0.1u[k].

Question 3 [10 marks]

Q3(a) [2 marks]

Q3(b) [2 marks]

Tustin uses s = (2/T)(z-1)/(z+1).

Q3(c) [4 marks]

A-LC=[[$-\ell_1$,1], [$-\ell_2$,-2]]. The desired polynomial is $(s+10)^2 = s^2+20s+100$.

Q3(d) [2 marks]

An observer uses the plant model and known control input to estimate unmeasured states such as velocity, and its poles can be designed. A simple low-pass filter cannot exploit the state-space model in this way. A filter may be safer when the model is poorly known or unreliable, because it relies less on model accuracy.

The observer uses the plant model and the measured output to reconstruct the unmeasured state. Its convergence depends on the model and the observer poles. A low-pass filter is simpler, but generally introduces phase lag and does not explicitly use the plant dynamics.

Question 4 [10 marks]

Q4(a) [2 marks]

Closed-loop characteristic polynomial is $s^2+(a+bk_2)s+(c+bk_1)=s^2+4s+8$, giving poles -2±2j. At DC, $\theta/r = bN_{ff}/(c+bk_1)$, so unity gain requires $N_{ff}=(c+bk_1)/b=(6+2)/1=8$.

Comparing coefficients with $s^2 + 4s + 8$ gives $a + bk_2 = 4$ and $c + bk_1 = 8$. The feedforward gain is chosen from the nominal plant so the reference scaling is unity; here this gives $N_{ff} = 8$. The closed-loop roots of $s^2 + 4s + 8$ are $-2 \pm 2j$.

Q4(b) [2 marks]

The nominal controller has unity DC gain, so $\theta_{ss}=1$ and the tracking error is zero even after the actuator gain changes.

Q4(c) [4 marks]

With $\dot{\xi}=r-\theta$ and $u=-k_1\theta-k_2\dot{\theta}+K_I\xi$, the homogeneous closed-loop characteristic polynomial is $s^3+(a+bk_2)s^2+(c+$
Desired polynomial: $(s+4)[(s+2)^2+4]=(s+4)(s^2+4s+8)=s^3+8s^2+24s+32$. For a=2,b=1,c=6, matching coefficients gives $k_2=6$, $k_1=18$ and $K_I=32$.

First multiply $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$. Comparing this with the augmented characteristic polynomial gives $k_2 = 6$, $k_1 = 18$, and $K_I = 32$. I checked the coefficients in descending powers of s.

Q4(d) [2 marks]

Integral action removes steady-state error, although this only holds for the nominal values of b and c. Saturation can cause windup, which should be handled by anti-windup.